

DEX Sentinel *Fix it before they call.*

Round 1 solution proposal · Outcome-based Digital Employee Experience (DEX) analytics · Working MVP · Vijay Vishwanath

1. PROBLEM STATEMENT

Correlate **subjective** employee-experience signals (service-desk calls, ticket language, frustration, repeat contacts) with **objective** endpoint telemetry (boot duration, application hangs, network quality, policy state, hardware health) to improve diagnosis, remediation and outcome-based reporting.

Why it matters today: the two signals live in different tools and are never joined; tickets close when calls stop, not when the laptop works; and the desk only learns about pain after the employee has lost hours. In our data, **46.8%** of contacts were repeats before a verified fix, slow boots (> 85 s) carried **1.9×** the frustration, and non-compliant devices raised login tickets in **28.5%** of weeks vs 0% when compliant.

2. PROPOSED SOLUTION: FOUR MOVES IN ONE LOOP

DEX Sentinel fuses what employees say with what their devices do, diagnoses the root cause, **predicts who will struggle next week**, and proves every fix with an outcome-based DEX Score.

Move	How it works	Output
1 · Fuse	Scores every ticket for frustration (explainable lexicon + repeat/escalation boost) and joins it to that device's telemetry for the same week.	Frustration Score 0–100, emotion, Device Health, correlation $r = 0.83$
2 · Diagnose	Fuses telemetry severity (65%) with ticket-language signal (35%) across 5 root-cause categories, then sub-cause rules and evidence; ML second opinion.	Ranked cause, likelihood, confidence, runbook fix, expected outcome
3 · Predict (ML)	LightGBM learns from telemetry trends, recent experience and device profile to forecast a frustrated ticket next week; calibrated and explained per device.	Proactive watchlist: risk %, drivers, fix, avoidable \$
4 · Prove	Measures before vs after for every remediation and rolls it into the DEX Score.	Experience Recovery %, repeat-contact drop, \$ value
+ DEX Copilot	Claude agent answering in plain English using 8 tools over the engines + RAG over 22 runbooks; offline grounded fallback.	Cited, number-grounded answers

3. AI, ANALYTICS AND SCORING APPROACH

- Frustration Score** = $\min(100, 8 + \sum \text{phrase weights} + 10 \times \text{prior contacts} + 15 \times \text{escalations})$; severity Low / Medium ≥ 40 / High ≥ 60 / Critical ≥ 80 .
- Device Health** = $0.8 \times (100 - 0.1 \cdot \text{boot} - 3 \cdot \text{hangs} - 4 \cdot \text{crashes} - 0.05 \cdot \text{latency}) + 0.2 \times \text{hardware health}$; **Telemetry Severity** = 60% worst + 40% mean of per-category breach severity vs warn/critical thresholds.
- Correlation engine:** Pearson + Spearman with p-values, severity lift by telemetry bucket, incidence lift for policy, impact ranking (excess tickets $\times$ frustration $\times$ repeat share).
- Predictive model:** LightGBM on 8 signals $\times$ (level, 1-week change, change vs 3-week mean, 4-week slope, weeks breached) + experience history + device profile; leakage-safe features, rolling out-of-time backtest vs the rule baseline, isotonic calibration, exact per-prediction driver contributions.
- DEX Score** = $0.35 \cdot \text{Experience Index} + 0.25 \cdot \text{Device Health} + 0.20 \cdot \text{Remediation Success} + 0.10 \cdot \text{Resolution Efficiency} + 0.10 \cdot \text{Sentiment Trend (fleet: 79.3, Good)}$. **Experience Recovery %** = $(\text{DEX after} - \text{before}) \div \text{before}$.

4. RESULTS FROM THE WORKING MVP

Evidence	Result	What it shows
Frustration vs telemetry correlation	$r = 0.83$ ($n = 326$)	What employees say reliably tracks what devices do
Frustrated tickets caught a week early (top 5% of devices)	72.3% ML vs 48.8% rules	Prediction is 1.48× better than today's rules; PR-AUC 0.455 vs 0.250
Calibration of predicted risk	25.6% predicted = 25.6% observed	Risk % can be trusted as a probability
Diagnosis example (TCK-00002)	App Crash · 100% / 99%	Text (62/100) + 9 hangs → fix KB-APP-001
Remediation outcomes (46 fixes)	Repeats 46.8% → 0.7%	45 of 46 improved; tickets/week 0.41 → 0.15
DEX Score recovery	64.0 → 87.6 (+36.9%)	Experience measurably recovered after fixes
Business value	\$114,734 / year	621 tickets avoided, 1,838 productive hours recovered

Prediction benchmark: 5,000 devices $\times$ 26 weeks, reproducible with seed 7; other figures from the bundled 260-device sample. All data is simulated.

5. ARCHITECTURE AND TECHNOLOGY

Raw exports (.xlsx/.csv, up to 200 MB) → ingestion with automatic column mapping and validation → in-memory analytical store (ticket ⇄ telemetry join) → six engines (experience, telemetry, correlation, diagnosis + ML, **prediction**, outcomes/DEX) → FastAPI (38 endpoints) → React UI (12 screens) and DEX Copilot. Uploads are validated off to the side and swapped in atomically; models retrain automatically after every upload. See the flow chart on page 3.

Layer	Technology
Backend	Python 3.11+, FastAPI, pandas, NumPy, SciPy, SQLAlchemy (SQLite / Postgres)
AI / ML	scikit-learn (TF-IDF + logistic regression, isotonic calibration), LightGBM, BM25 RAG, Claude (claude-opus-5-5) or Azure OpenAI
Frontend	React 18, TypeScript, Vite, TanStack Query, Recharts
Delivery	Docker (single container), Google Cloud Run, one-command run scripts; 81 automated tests

6. INNOVATION, RESPONSIBLE AI AND FEASIBILITY

- **Predictive DEX:** moves the service desk from reactive to proactive: flags employees before they call, explains why, and prices the fix (DEV-2462 was flagged at 38.6% risk a week before an "I'm losing hours" ticket).
- **Explainable and measured:** every score traces to phrases, thresholds or feature contributions; ML is always shown next to the rule baseline it must beat, on unseen weeks.
- **Grounded AI:** the LLM can only quote numbers returned by tools and cites runbooks; the whole app runs offline without an AI key; names can be pseudonymised.
- **Feasible now:** works on standard ServiceNow / Intune / Nexthink exports; 163 MB (2.34 M device-weeks) analysed in about 80 s; deployable as one container.

7. BUSINESS IMPACT

Lever	Measured value	How it is calculated
Proven value of past fixes	\$114,734 / year	Tickets avoided × \$22 + resolution hours × 50% productivity loss × \$55/h
Proactive value (top 50 devices)	≈ \$3,699 / week	Risk × historical ticket-rate reduction of the fix × cost per ticket
Repeat contacts	46.8% → 0.7%	Fewer call-backs once fixes are verified by outcome
Productive time	1,838 hours / year	Hours employees get back from avoided tickets

All cost assumptions are editable in the app's Settings, and every value recalculates.

8. HOW DEX SENTINEL MEETS THE EVALUATION CRITERIA

Criterion	Evidence
Innovation	Predictive DEX: forecasts frustration from telemetry trends before the call, explained per device and priced
Technical depth	Fusion root-cause engine, leakage-safe ML with out-of-time backtest and calibration, agentic LLM with RAG
Impact	72% vs 49% of frustrated tickets caught early; +36.9% experience recovery; \$115k/yr proven value
Feasibility	Runs on standard exports, 163 MB in ~80 s, single container, fully offline mode
User experience	Dashboard → watchlist → device → diagnosis → Copilot in a few clicks; every chart has a table view
Completeness	Working MVP: 12 screens, 38 APIs, 81 automated tests, full documentation and pitch deck

9. LIVE DEMO SCENARIO (5 MINUTES)

- **Executive Dashboard:** DEX Score 79.3 and the insight "185 devices likely to raise a frustrated ticket next week".
- **Proactive Watchlist:** ML vs rules (72% vs 49%), ranked devices with drivers, fix and avoidable value.
- **Device 360, DEV-2462:** boot time climbs 36 → 109 s; risk rises to 38.6% a week before the employee's first call.
- **Diagnosis Assist and Copilot:** one-click root cause with evidence, then a cited plain-English answer.
- **Outcome Reporting:** 45 of 46 fixes improved; DEX 64.0 → 87.6.

10. ROADMAP AND ASK

Next: pilot on one business unit's real exports (the built-in backtest reports real accuracy in week 1) → live ServiceNow / Intune / Nexthink connectors → multilingual AI text understanding → one-click proactive remediation with human approval. **Ask:** a pilot with one business unit.

DEX Sentinel: solution flow chart

From raw exports to proven outcomes. The proactive path (new) predicts problems before the employee calls; the reactive path diagnoses a ticket when it arrives. Both end in a verified fix whose impact is measured and fed back.

